Battle of the Bands: Trying to Identify Contributions to a Collaboration

CAUSE Webinar

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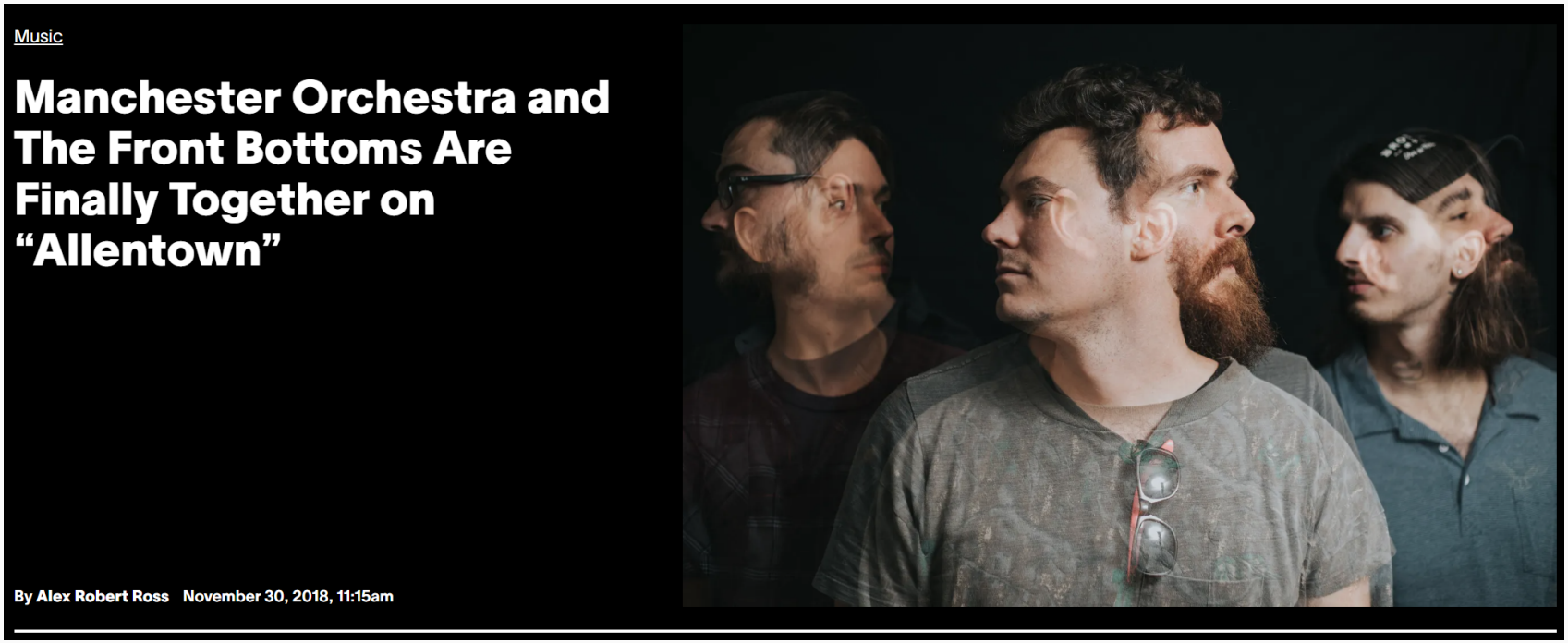
The Goal

- It is hard to find real world data sets that work for teaching at a variety of levels.
- We need data that is clean enough for students, yet messy enough to show realities of working with data.
- Bonus points if the data can be used for various models or applications so students can re-visit it and see how much they have learned!

Our Application



Source: Ross (2018)

Our Application

- Three Indie rock bands collaborated to produce a single track, "Allentown".
- Can we use statistics and data science methods to determine which of the 3 bands the collaborative track:
  - Most sounds like?
  - Has lyrics the most similar to?

The Problem (and the opportunity!)

Problem

- To answer our research questions, we need a data set with information about the sound qualities of the combined track and the tracks of the 3 bands in general.
- We also need information about the lyrics for all of these tracks.
- However, no such data set exists!

Opportunity

- We create activities to show students at a variety of levels how to (1) create the needed data, (2) clean and process it, and (3) use modeling to answer our research questions.

Where we teach

Will

- Colgate University
- Courses Used:
- Students per class: 30 or less

Roy

- Furman University
- Courses Used:
- Students per class: 30 or less

Ciaran and Nicole

- Wake Forest University : medium-sized liberal arts college
- Courses Used: Graduate GLMS, Undergraduate Text Analysis
- Students per class: 30 or less

What we have: Tracks

- **The Front Bottoms Releases (61 tracks):** The Front Bottoms; Talon of the Hawk; Rose; Back on Top; Needy When I'm Needy; Going Grey; Ann; and End of Summer (Now I Know)
- **Manchester Orchestra Releases (77 tracks):** You Brainstorm, I Brainstorm, but Brilliance Needs a Good Editor; I'm Like a Virgin Losing a Child; Fourteen Years of Excellence; Mean Everything to Nothing; Simple Math; Cope (Deluxe Version); Hope; A Black Mile to the Surface
- **All Get Out Releases (42 tracks):** All Get Out; The Season; Movement; Nobody Likes a Quitter; No Bouquet

What we have: Lyrics

[redacted]

⋮

Teaching Opportunity 1: Creating Usable Data

- While we have all these tracks, we do not have information about the sound or the lyrics in a way that we can use for modeling.
- How do we get it??

Where the students minds likely go...

- AI!
- However, there are legal and ethical issues with using AI to create features from copyrighted music.
- Example: Genius for lyrics (ethical issues) and Spotify AI (no longer allowed due to AI)
- Storytime: What literally happened as we were writing the paper...

Opportunity

- All of this gives a rich space for discussion about limitations and ethical considerations with AI.
- It also gives an opportunity to show students alternatives to AI that avoid these pitfalls.

Our solution for sound features: Essentia

- We use Essentia (Bogdanov et al. 2013; Alonso-Jiménez et al. 2020) to extract meaningful information from the spectrograms of each track.
  - **Music Extractor:** loudness, silence rates, beats per minute, danceability, key, mode, etc. as extracted from the spectrogram. (67 variables)
  - **Models:** approachability, engagement, arousal, valence, happy, sad, etc. as predicted using models built from existing databases. (14 variables)

Our solution for sound features: LIWC

- We use the Linguistic Inquiry and Word Count (LIWC or 'Luke') (Boyd et al. 2022) to use psychometric dictionaries that provide insight into a writer's psychology, social concerns, or writing style (112 variables).
- This is legal to use for research purposes and work with students.

The Result

- A large data set with many variables
- An outcome variable with 3 levels
- A lot of opportunities for student engagement
- A great space for student discussion about AI, data creation, and data privacy.

Teaching Opportunity 2: Cleaning and Processing

The data set requires:

- Handling missing data.
- Transformation of features.
- Data tidying like name and variable type changes.
- Handling duplicate tracks (same lyrics, different sounds!)

Teaching Opportunity 2: Cleaning and Processing

- Educators can choose how clean or messy to make the data based on how many cleaning steps are pre-processed.
- This makes the data set a good option for a variety of learning goals.

Teaching Opportunity 3: Modeling

Logistic Regression

- Only two of the three bands have performing credits, so it is reasonable to treat $Y =$ performing artist as binary

Multinomial Regression

- All three artists have writing credit, so we can treat $Y =$ writing artist as a 3 level variable.

What we created: Shiny Apps

Apps: https://shiny.colgate.edu/apps.html

- **Exploratory Data Analysis (EDA)**
- **t-test** or **ANOVA** (For comparing bands)
  - non-parametric alternatives (Mood's Median, Kruskal Wallis)
- **Logistic Regression**
- **Multinomial Regression**

What we created

We created a variety of modeling activities for students in intro level regression courses through graduate courses, including:

- Logistic regression
- Multinomial regression
- Cross validation to assess predictive ability
- Model diagnostics (ex: quantile residuals)
- Feature selection (ex: lasso)
- Classification trees

Example Activity: Residuals for Logistic Regression

- Randomized Quantile Residuals (Dunn and Smyth 1996)

$$a_i = \begin{cases} 0 & y_i = 0 \\ 1 - \hat{\pi}_i & y_i = 1 \end{cases}$$

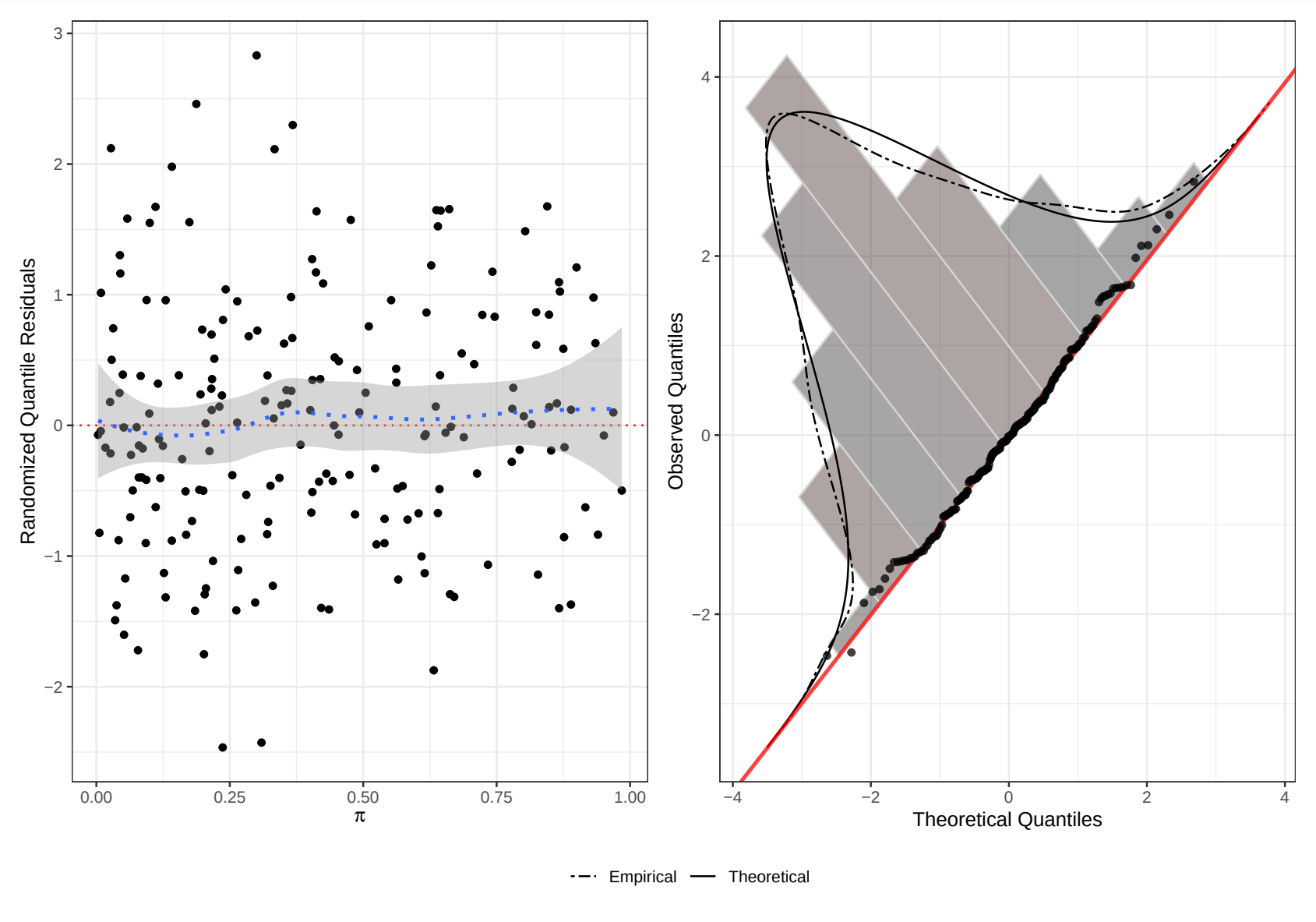
$$b_i = \begin{cases} 1 - \hat{\pi}_i & y_i = 0 \\ 1 & y_i = 1 \end{cases}$$

$$u_i \sim \text{random uniform}(a_i, b_i)$$

$$e_i = \Phi^{-1}(u_i)$$

where $\Phi^{-1}(\cdot)$ is the inverse standard Gaussian CDF.

Good


Randomized Quantile Residuals
π
0.00
0.25
0.50
0.75
1.00
3
2
1
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Observed Quantiles
Theoretical Quantiles
−4
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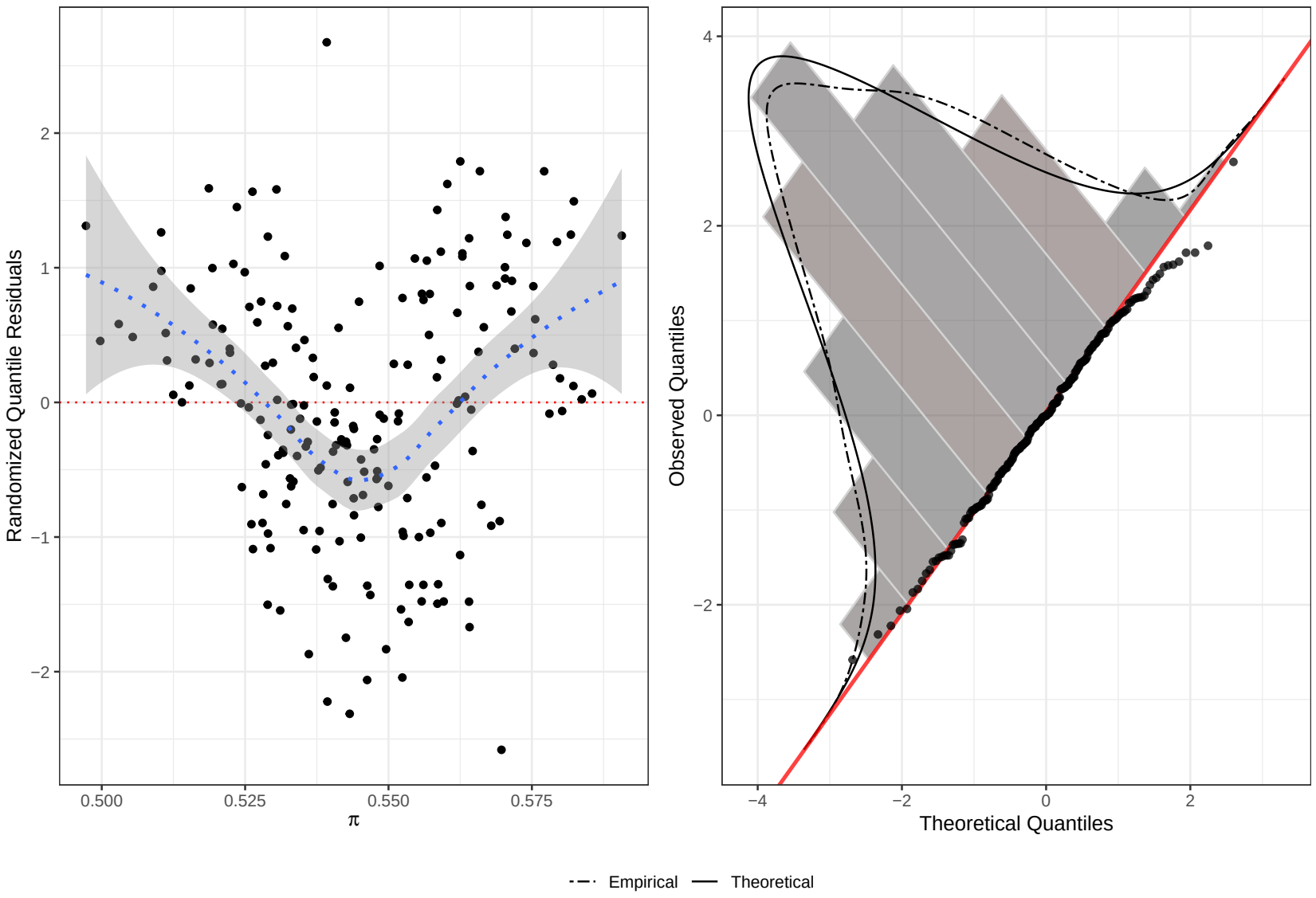
0
2
4
Empirical
Theoretical

Bad


Randomized Quantile Residuals
2
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−1
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0.500
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0.550
0.575
π
Observed Quantiles
4
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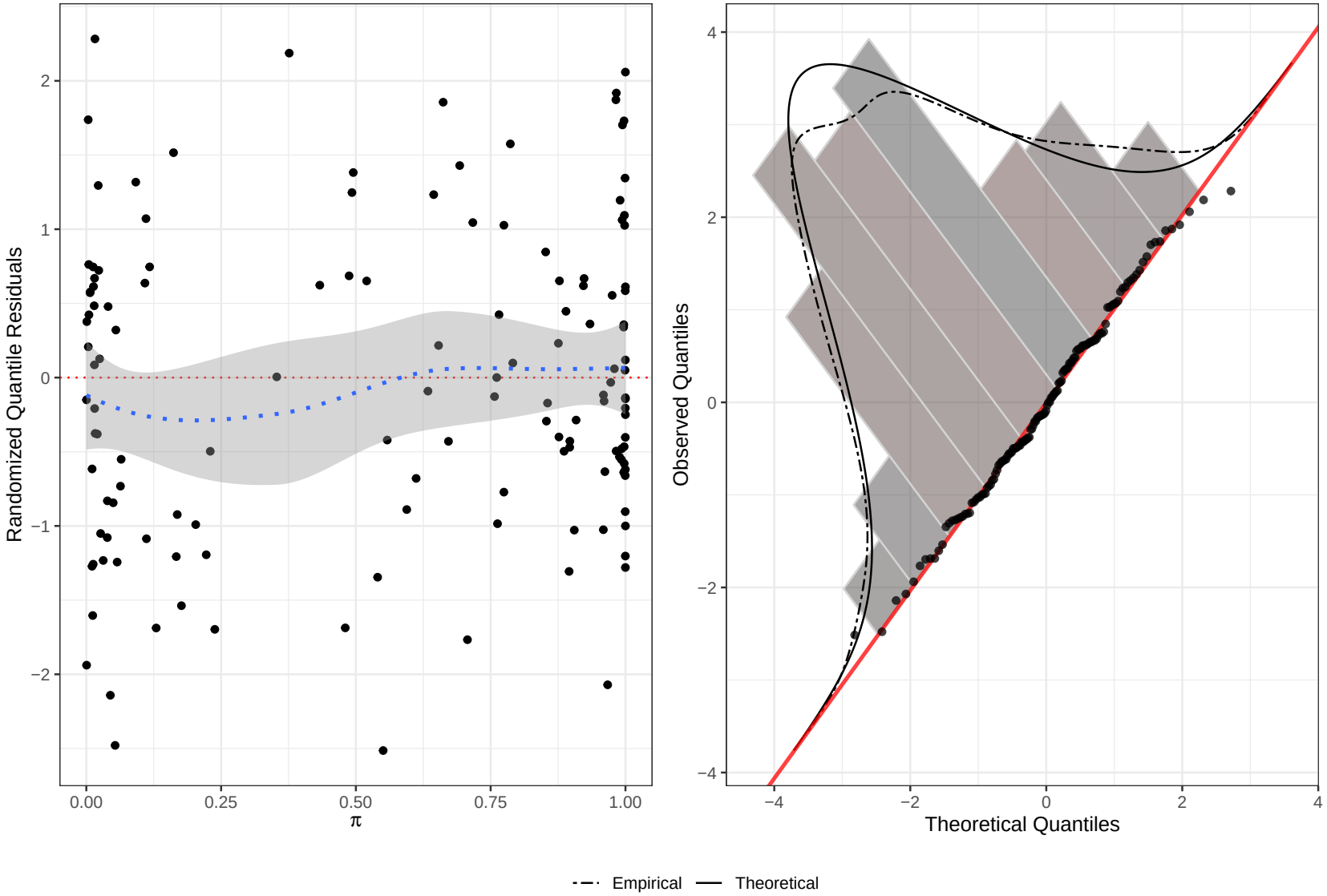
2
Theoretical Quantiles
Empirical
Theoretical

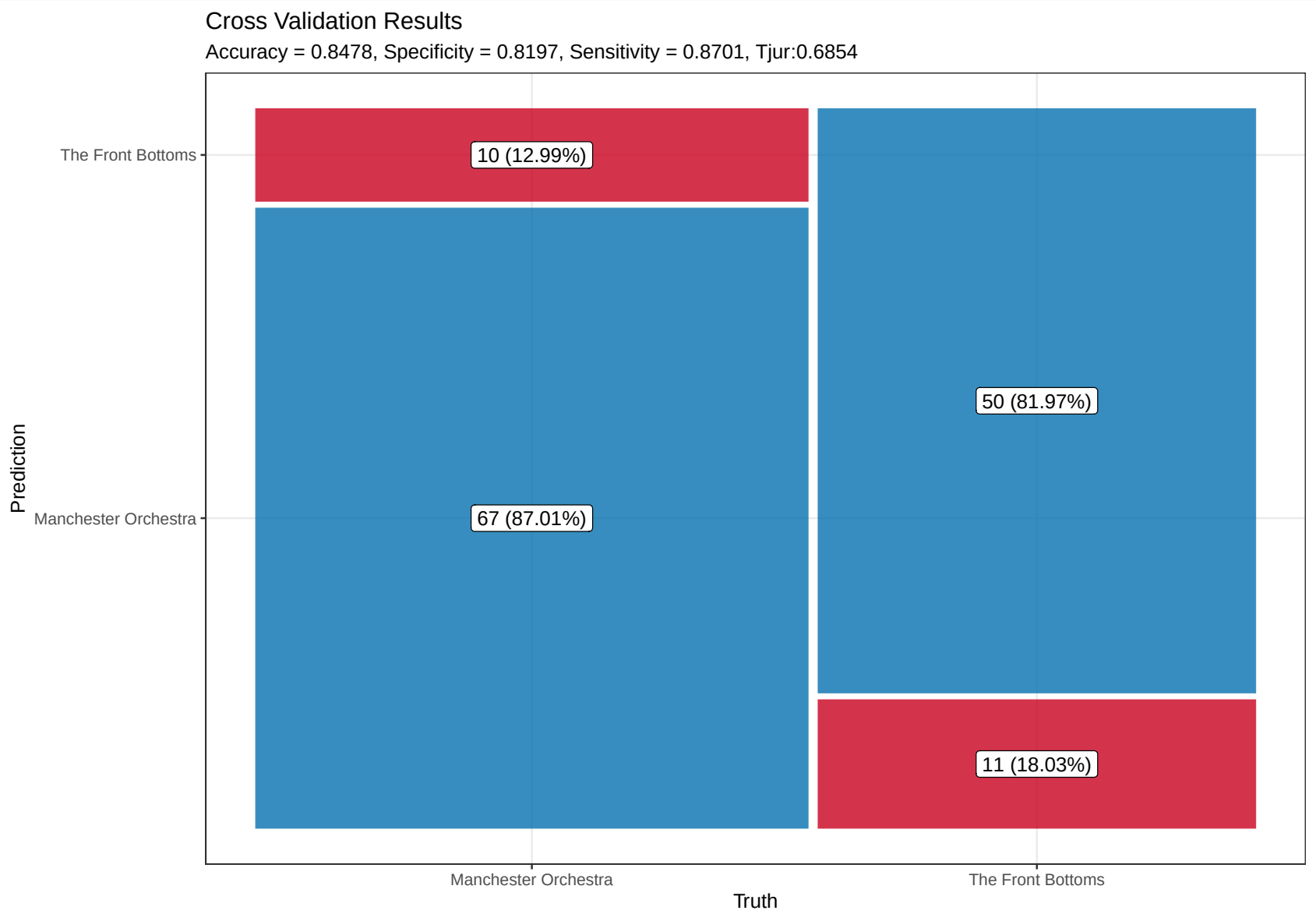
Sound Logistic Model (Manchester Orchestra v. The Front Bottoms)



Results (Sound Model)

	Estimate	SE	Z	p-value
(Intercept)	1.10	0.45	2.44	.01
overall_loudness	-1.83	1.26	-1.45	0.15
spectral_energy	-1.27	0.57	-2.24	.02
pitch_salience	0.35	0.50	0.69	0.49
tempo	0.37	0.40	0.92	0.36
dissonance	-2.48	1.21	-2.05	.04
danceability	0.43	0.44	0.98	0.33
tuning_frequency	1.06	0.43	2.45	.01
duration	0.79	0.40	1.96	.05
valence	-0.57	1.18	-0.49	0.63
arousal	0.28	0.78	0.35	0.72
acoustic	-1.73	0.72	-2.40	.02
timbreBright	1.56	0.48	3.23	$<.0001$
approachability	-1.43	0.70	-2.06	.04
engagement	-1.07	0.54	-1.98	.05

Results (Sound Model)



So, did we get it right? Interview with the Artists

- Andy Hull of Manchester Orchestra worked out the melody and music.
- Brian Sella of The Front Bottoms then helped develop the chorus.
- This means it makes sense that by the sound, the model would predict Manchester Orchestra!

What about the lyrics??

If we want to model the lyrics, we now have all three bands in play, so Y is multinomial.

- Using our features from LIWC, can we predict who the lyrics most "read like"??
- For this, we use multinomial lasso.

Results (Lyric Model): 120 features reduced to 29

term	FB	MO	AGO	term	FB	MO	AGO
WC	.0009	-.0013	.004	relig	-0.1916	0.1087	.0829
Tone	.005	-.057	.0007	want	-.0081	.0044	.0037
fnct	.0006	.007	-.0076	curiosity	-.0493	.0816	-.0324
you	-.0085	.0009	.0076	allure	-.0167	-.0224	.0391
shehe	.0062	.0033	-.0095	visual	-.0686	0.1385	-.0699
conj	.0703	.0307	-.1011	auditory	-.0006	.0273	-.0267
negate	-.0040	.0025	.0015	feeling	.1063	-.0995	-.0068
quantity	-.0239	-.0561	.0800	focuspast	-.0582	.0795	-.0213
achieve	-.0160	.0150	.0010	netspeak	.0985	-.0537	-.0448
certitude	-.0386	.0342	.0045	Comma	.0151	-.0038	-.0113
emotion	.0143	-.0162	.0019	Apostro	-.0041	.0001	.0040
Culture	.3465	-.1260	-.2205	OtherP	.0512	-.0269	-.0243
politic	-.4229	-.0675	.4904	positivewords	.0031	-.0043	.0012
tech	.0529	-.0184	-.0345	negativewords	.0055	-.0263	.0208
leisure	.0922	-.0287	-.0635				

Teaching Opportunity: Talking to a Client

How do we break all of this down for a client??

- Manchester Orchestra has the most negative lyrics (`Tone`) and the least emotion (`emotion`).
- The Front Bottoms tend to use variants of "you" the least (`you`) but have lyrics about tech (`tech`) more than the other 2 groups.
- All Get Out uses the most quantity words, as well as the most words about politics.

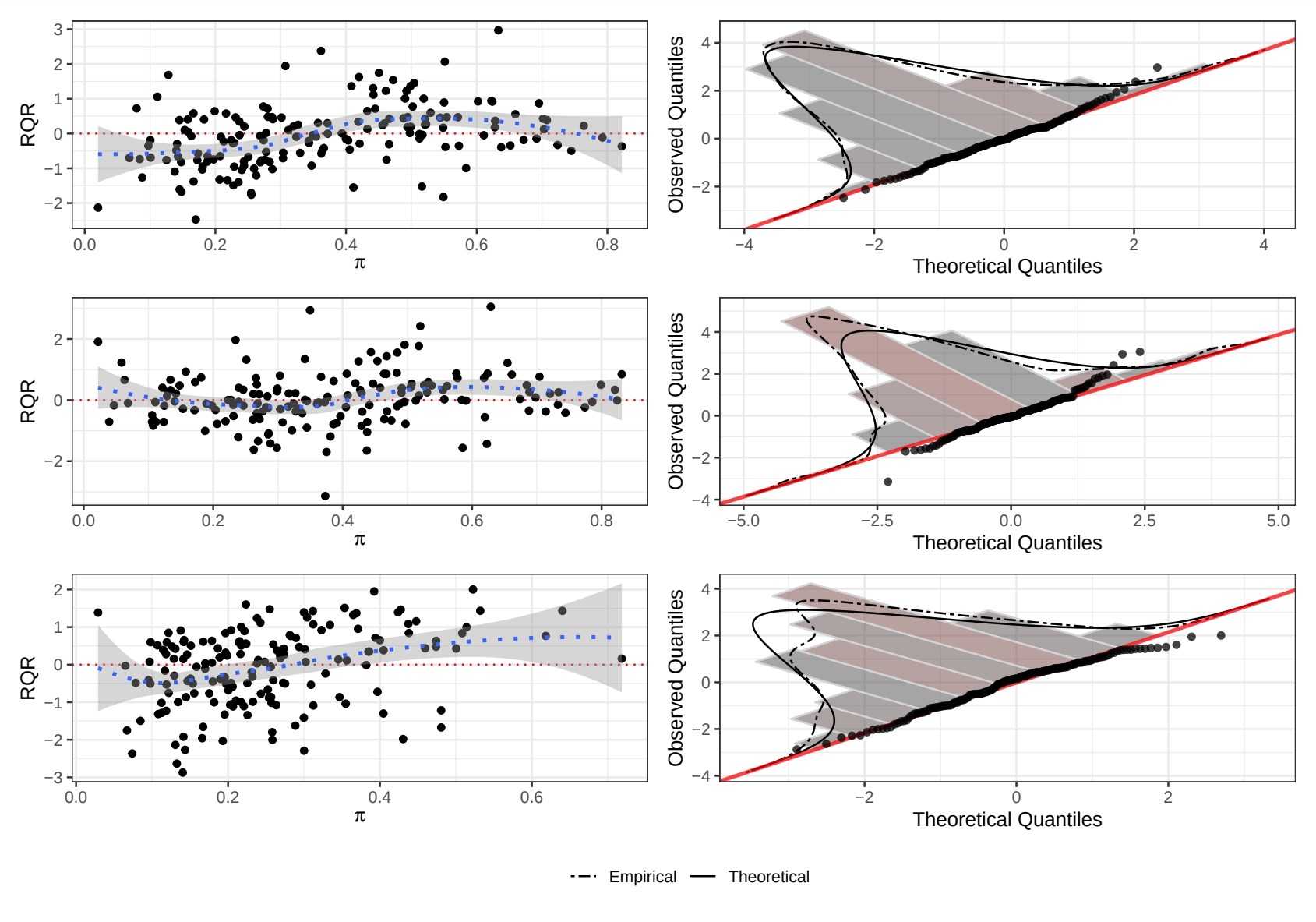
And so on. This gives a nice opportunity for students to use data dictionaires to translate modeling results into concrete findings.

Results (Lyric Model):

What some of this means

term	FB	MO	AGO	term	FB	MO	AGO
WC	.00	-.00	.00	relig	-0.19	0.11	.08
Tone	.00	-.01	.00	want	-.01	.00	.00
fnct	.00	.01	-.01	curiosity	-.05	.08	-.03
you	-.01	.00	.01	allure	-.02	-.02	.04
shehe	.01	.00	-.01	visual	-.07	0.14	-.07
conj	.07	.03	-0.10	auditory	-.00	.03	-.03
negate	-.00	.00	.00	feeling	0.11	-0.10	-.01
quantity	-.02	-.06	.08	focuspast	-.06	.08	-.02
achieve	-.02	.01	.00	netspeak	0.10	-.05	-.04
certitude	-.04	.03	.00	Comma	.02	-.00	-.01
emotion	.01	-.02	.00	Apostro	-.00	.00	.00
Culture	0.35	-0.13	-0.22	OtherP	.05	-.03	-.02
politic	-0.42	-.07	0.49	positivewords	.00	-.00	.00
tech	.05	-.02	-.03	negativewords	.01	-.03	.02
leisure	.09	-.03	-.06				

Results (Lyric Model)



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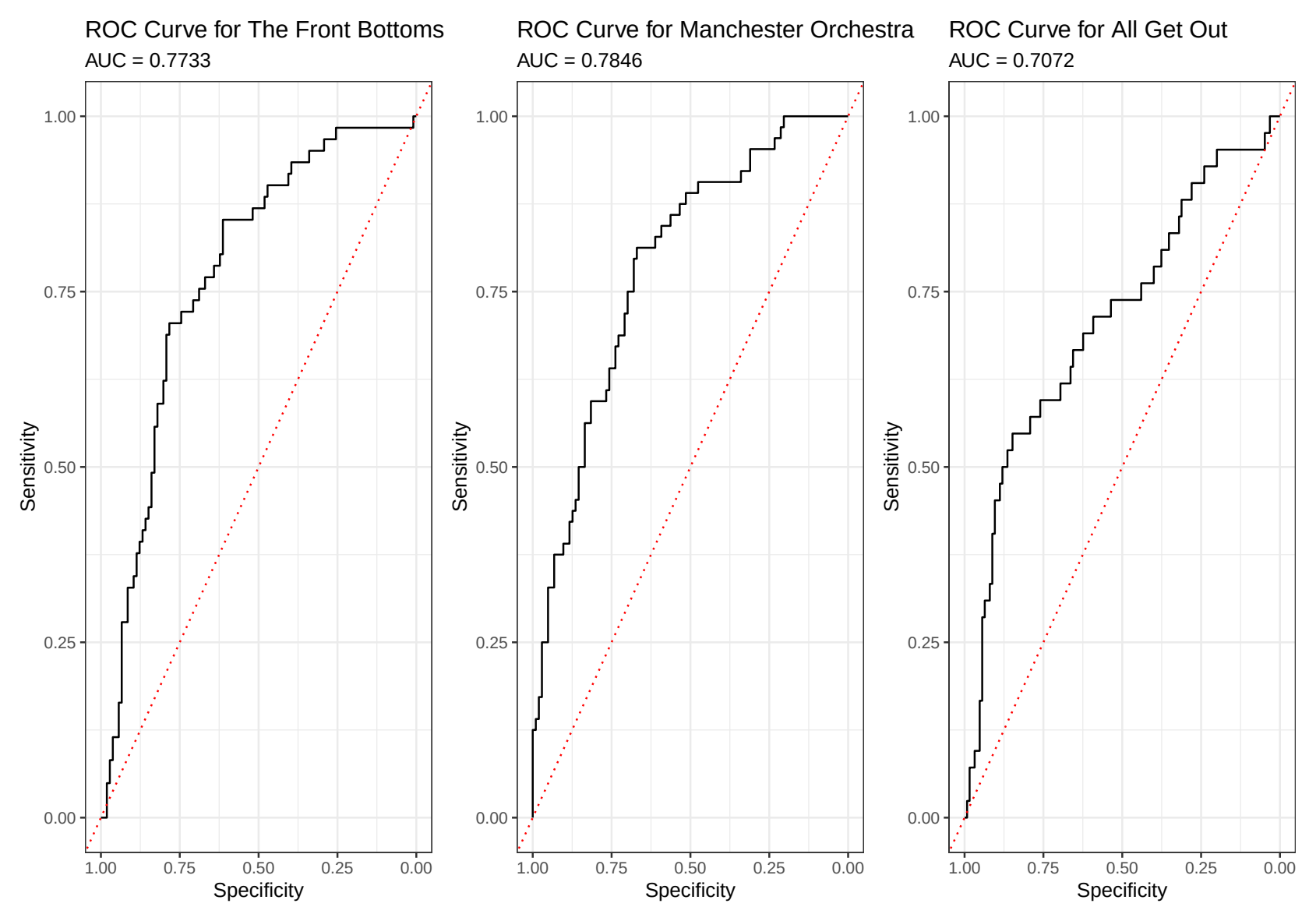

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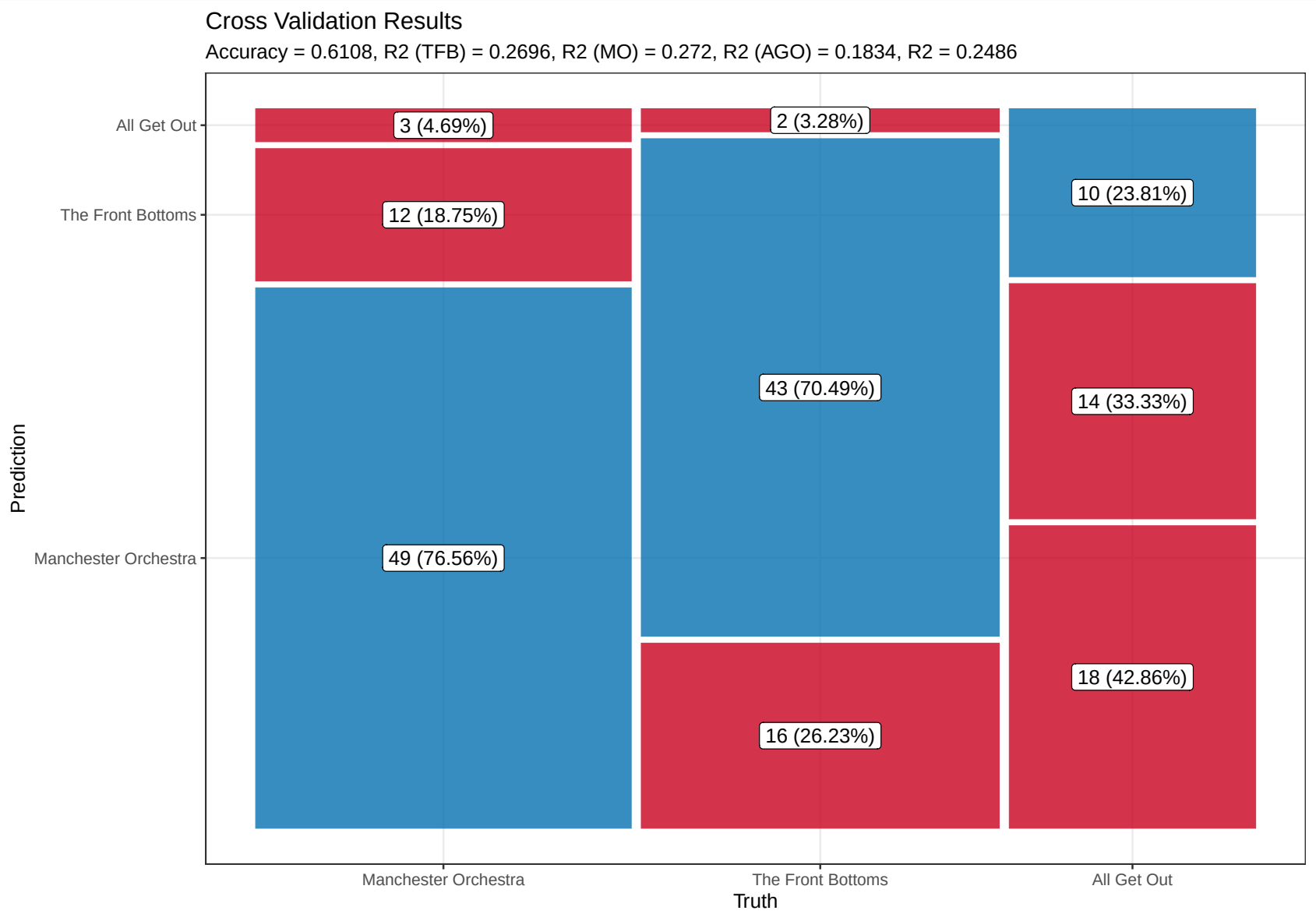

Results (Lyrics Model)

- The condition for using the multinomial model are met.
- The model fits well.
- The model predicts that Allentown is a All Get Out song by the lyrics!

$$P(Y = \text{The Front Bottoms}) = 0.1709$$
$$P(Y = \text{Manchester Orchestra}) = 0.2760$$
$$P(Y = \text{All Get Out}) = 0.5531$$

- However, the model is less sure that when we will modeling the "sounds like" outcome.
- Does this make sense?

Reality

- The creation of this track started when Nate Hussey of All Get Out sent the first four lines of the track.
- Andy Hull of Manchester Orchestra worked out the melody and music.
- Brian Sella of The Front Bottoms then helped develop the chorus.
- This means that is makes sense that All Get Out has the largest probability, then Manchester Orchestra, and then The Front Bottoms!

Thank You!

Questions?

References

Alonso-Jiménez, Pablo, Dmitry Bogdanov, Jordi Pons, and Xavier Serra. 2020. "Tensorflow Audio Models in Essentia." *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 266–70.

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Boyd, Ryan L, Ashwini Ashokkumar, Sarah Seraj, and James W Pennebaker. 2022. "The Development and Psychometric Prop-